

```
probe description
/predicate/
{
```

```
    actions
```

```
}
```

When the D script is executed, the probes described in the probe description are enabled. The action statements are executed when the probe fires (the probe event occurs) and the predicate evaluates to true.

Consider the following simple D script:

```
syscall::write:entry
/execname == "bash"/
{
```

```
    printf("bash with pid %d called write \
          system call\n",pid);
```

```
}
```

In this D script, the probe description is `syscall::write:entry` which describes the write system call.

The predicate is `/execname == "bash"/`. This script checks whether the executable that is calling the write system call is the bash shell. The `printf` statement is the action that executes every time bash calls the write system call.

Predicates and action statements are optional.

- If the predicate is missing, then the action is always executed.
- If the action is missing, then the name of the probe that fired is printed.

In the following simple D script the predicate is missing, so the action is always executed. This script prints all the processes successfully started in the system.

```
proc:::exec-success
{
```

```
    trace(execname)
```

```
}
```

The following D script is more complex. This script shows all the files that are opened in the system.

```
syscall::open:entry
{
```

```
    printf("%s opened %s\n",execname,copyinstr(
```

Quick Tip

You need to be root (or a user privileged with DTrace rights in RBAC). You can be root by executing `$ pfexec su -`